Supplementary material

How much is the wind worth? Sprints and horizontal jumps from age 10 to senior level

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Supplementary Methods

Sample construction: duplicate records

Before the filters described in the main text (Section 2.1), we removed duplicate records: 47 962 board-jump rows that re-imports in 2026 had copied from an identical take-off-zone result (same athlete, date, mark and wind; almost all at ages 11–13, and listed in the source only under the zone event), 8 907 such copies from multi-day meets, dated to the meet's first day while the zone result lies one to six days later, and 770 exact duplicates, mostly meets stored twice. Because the 2026 import filed take-off-zone jumps under the board events, the 1 597 board-event jumps at ages 11–13 in 2026 were also excluded.

Short hurdles

The hurdles extension covers the ten short-hurdle event codes with substantive wind data (60 m to 110 m hurdles; each code specifies the class's hurdle height, so a code identifies an age-class specification), with the filters and estimator of the main events; the hurdles extension contributed 33 205 analyzed rows. It was added after the main estimations by a post-plan decision.

In the age-class system, hurdle events carry class-specific heights, so each event code identifies an age-class specification. Across the ten codes with wind data (Table S6; Figure S2), the 60 m hurdles matched the flat 60 m of the same sex and class: 0.055–0.067 s per m/s against 0.056–0.070, point estimates from 21 % below to 9 % above the flat references, with overlapping intervals. Over 100 and 110 m the hurdle estimates mostly exceeded the flat 100 m reference, from 4 % below to 48 % above; in the girls' 76.2 cm class (0.108 against 0.082) the hurdle interval lies entirely above the flat interval, and in the boys' 84 cm class (0.133 against 0.090) it lies above the flat point estimate but overlaps the flat interval. Per second of race time the pooled hurdle codes (0.0048–0.0069 s per m/s) span the straight-sprint rates (Table S4), so the hurdles' longer race times account for much of the difference, though not all of it in those two youth classes. Within the 100 m hurdles the youth codes also exceeded the senior code (0.108 s per m/s for girls' 76.2 cm at 16–17 and 0.133 for boys' 84 cm at 14–15, against 0.069 for the senior women's 84 cm event), although these comparisons cross sexes and hurdle heights.

Sample size: sensitivity analysis

Because the sample is the eligible population (main text, Section 2.3), we report an a posteriori sensitivity analysis for the primary outcome, the wind slope in raw units (s or cm per m/s): the minimum detectable effect (MDE) of a two-sided Wald test of the meet-clustered slope at $\alpha = 0.05$ with power 0.80, MDE = (1.96 + 0.84) $\times$ SE, computed in Python from the estimated standard errors. The smallest effect size of interest is the slope at which a legal +2.0 m/s tailwind moves a mark by one recording unit: 0.005 s per m/s in the sprints, which are timed to 0.01 s, and 0.5 cm per m/s in the jumps, measured to 1 cm. At event level the MDE was 0.003 and 0.004 s per m/s in the 60 and 100 m and 0.4 and 0.3 cm in the long jump and its zone variant, all below that threshold, against 0.013 s in the 200 m, 0.018 and 0.053 s in the small 80 and 150 m events, and 0.9 and 1.4 cm in the two triple jumps. In the sex $\times$ age-band sprint models (main text, Table 3) it ranged from 0.004 to 0.086 s per m/s (median 0.014), largest for the 200 m at 10–11, one of the thin samples marked in Table 4. To protect precision, estimates are reported only where at least 300 results in 100 athlete-seasons contribute, and standard errors are clustered on meet, which gives wider intervals than the athlete clustering of the analysis plan. No participants were recruited, so no adjustment for attrition applies.

Departures from the analysis plan

Besides the post-plan checks and extensions named in the main text, the analysis departed from the plan as follows. H5 is valued from the 0 to +2 m/s slope against a robust scale and for the sprints only (the plan named the +2.0 m/s advantage against the classical within-cell standard deviation, for all events); ability is the wind-adjusted rather than the raw season mean; the planned formal test of the age gradient (H2) was not run, and the age bands are reported with their intervals instead; the linear model uses readings within $\pm$8 rather than $\pm$5 m/s; and the drag-only benchmark, the per-second scaling, meet fixed effects by age band, the age-graded points, the 2025 lists, the sex- and age-specific asymmetry comparison (Table S8) and the external-sample comparison (Table S3) were added after the plan.

Supplementary tables

Table S1: Robustness: gain per +1 m/s under alternative specifications (point estimates; s for sprints, cm for jumps).

Event	Primary	Cells ≥3	No exact 0.0	Excl. ±0.2 of +2.0	2011–19	2021–26	Meet FE
60 m	0.0591	0.0590	0.0591	0.0587	0.0600	0.0573	0.0513
80 m	0.0746	0.0711	0.0761	0.0729	0.0740	0.0779	0.0768
100 m	0.0823	0.0816	0.0823	0.0829	0.0843	0.0795	0.0659
150 m	0.0814	—	0.0780	0.0850	—	—	0.0838
200 m	0.1215	0.1203	0.1215	0.1212	0.1258	0.1135	0.0713
Long jump	3.1597	3.1129	3.2467	3.1932	3.2987	2.9343	4.1017
Long jump (zone)	2.1133	2.0471	2.1538	2.0984	2.1337	2.1518	2.6624
Triple jump	4.1848	4.1647	4.2642	4.1691	4.0189	4.7732	6.1704
Triple jump (zone)	3.8821	3.3154	3.8589	3.7725	3.7075	4.8247	5.0930

Point estimates are identical under athlete and meet clustering; only intervals differ. The 150 m lacks the cells-≥3 and period columns for want of data.

Table S2: Asymmetry: separate tail- and headwind slopes, observed head/tail ratio, and the ratio drag-only physics predicts for the same estimator (95 % CIs clustered on meet).

Event	Unit	Tailwind gain per +1 m/s	Headwind loss per −1 m/s	Ratio	Drag-only ratio	p equal slopes
60 m	s	0.047 (0.044–0.051)	0.072 (0.068–0.076)	1.52	1.84	<0.001
80 m	s	0.072 (0.048–0.097)	0.078 (0.054–0.101)	1.07	1.72	0.805
100 m	s	0.065 (0.061–0.068)	0.107 (0.102–0.113)	1.66	1.72	<0.001
150 m	s	0.093 (0.031–0.154)	0.067 (−0.031–0.164)	0.72	1.80	0.721
200 m	s	0.078 (0.067–0.089)	0.184 (0.168–0.199)	2.36	1.85	<0.001
Long jump	cm	2.93 (2.58–3.28)	3.71 (2.77–4.65)	1.27	—	0.177
Long jump (zone)	cm	2.48 (2.19–2.77)	1.41 (0.95–1.87)	0.57	—	<0.001
Triple jump	cm	4.63 (3.78–5.48)	2.79 (0.78–4.80)	0.60	—	0.136
Triple jump (zone)	cm	4.69 (3.40–5.98)	1.90 (−0.25–4.04)	0.40	—	0.044

Drag-only ratio: time changes proportional to the change in squared airspeed at the athlete's mean speed, scaled to the observed linear slope, imposed on the observed winds and estimated with the same piecewise model. The simpler ratio $(2v + 1)/(2v - 1)$ for a wind of exactly ±1 m/s is 1.14–1.18. No benchmark is given for the jumps, where wind acts on run-up and flight differently.

Table S3: The elite end with the same estimator: wind effects in US collegiate results (TFRRS) and in the career results of championship finalists (Tilastopaja), with the registry's all-age and senior rows as reference (within athlete × season; 95 % CIs and SEs clustered on meet).

Sample	n	Mean mark	Per m/s, linear (95 % CI)	Per s	Tailwind	Headwind	Head/tail (drag)	0 to +2	Above +2 (SE)	Meet FE
100 m, men										
Registry, all ages	34 165	12.40	0.080 (0.077–0.083)	0.0064	0.066	0.101	1.54 (1.67)	0.073	0.049 (0.005)	0.064
Registry, 20–34	8 909	11.42	0.071 (0.066–0.075)	0.0062	0.064	0.082	1.27 (1.58)	0.073	0.047 (0.007)	0.060
US collegiate (TFRRS)	57 534	10.89	0.054 (0.053–0.056)	0.0050	0.047	0.077	1.62 (1.67)	0.051	0.043 (0.002)	0.045
Championship finalists (Tilastopaja)	8 885	10.19	0.048 (0.045–0.051)	0.0047	0.036	0.070	1.93 (1.45)	0.038	0.033 (0.005)	0.040
100 m, women										
Registry, all ages	30 392	13.84	0.085 (0.082–0.089)	0.0062	0.065	0.114	1.77 (1.73)	0.067	0.059 (0.006)	0.071
Registry, 20–34	4 543	12.63	0.072 (0.067–0.078)	0.0057	0.057	0.098	1.71 (1.58)	0.055	0.063 (0.008)	0.062
US collegiate (TFRRS)	63 234	12.22	0.058 (0.056–0.060)	0.0047	0.050	0.081	1.61 (1.76)	0.059	0.041 (0.002)	0.047
Championship finalists (Tilastopaja)	8 047	11.25	0.059 (0.055–0.063)	0.0052	0.044	0.083	1.86 (1.49)	0.047	0.039 (0.008)	0.053
200 m, men										
Registry, all ages	29 352	27.07	0.117 (0.108–0.127)	0.0043	0.071	0.183	2.56 (1.82)	0.081	0.053 (0.015)	0.067
Registry, 20–34	4 931	23.08	0.085 (0.068–0.101)	0.0037	0.048	0.151	3.15 (1.64)	0.070	0.013 (0.027)	0.053
US collegiate (TFRRS)	57 111	22.05	0.066 (0.062–0.069)	0.0030	0.052	0.104	2.00 (1.64)	0.057	0.045 (0.005)	0.050
Championship finalists (Tilastopaja)	6 265	20.54	0.065 (0.057–0.074)	0.0032	0.030	0.114	3.74 (1.41)	0.034	0.021 (0.016)	0.068
200 m, women										
Registry, all ages	29 894	29.86	0.125 (0.115–0.136)	0.0042	0.083	0.185	2.22 (1.87)	0.110	0.033 (0.015)	0.077
Registry, 20–34	2 954	25.83	0.095 (0.075–0.115)	0.0037	0.047	0.180	3.81 (1.76)	0.071	0.011 (0.029)	0.059
US collegiate (TFRRS)	71 467	25.30	0.074 (0.070–0.078)	0.0029	0.057	0.118	2.06 (1.81)	0.072	0.040 (0.005)	0.056
Championship finalists (Tilastopaja)	5 804	22.93	0.078 (0.069–0.088)	0.0034	0.055	0.110	2.00 (1.50)	0.063	0.037 (0.022)	0.079
100 m hurdles, women										
US collegiate (TFRRS)	61 051	14.74	0.075 (0.072–0.079)	0.0051	0.057	0.129	2.27	0.072	0.038 (0.005)	0.062
Championship finalists (Tilastopaja)	7 302	12.94	0.054 (0.049–0.059)	0.0041	0.033	0.084	2.51	0.031	0.039 (0.010)	0.045
110 m hurdles, men										
US collegiate (TFRRS)	45 225	14.97	0.057 (0.054–0.060)	0.0038	0.043	0.099	2.31	0.051	0.034 (0.004)	0.043
Championship finalists (Tilastopaja)	6 788	13.47	0.044 (0.039–0.050)	0.0033	0.026	0.072	2.77	0.027	0.023 (0.009)	0.046
Long jump, men										
Registry, all ages	15 259	4.69	3.34 (2.90–3.78)	—	2.98	4.27	1.43	1.54	5.73 (0.60)	4.77
Registry, 20–34	1 813	6.26	3.11 (1.69–4.52)	—	2.85	4.17	1.46	2.70	3.11 (1.71)	8.56
US collegiate (TFRRS)	38 335	6.77	3.74 (3.46–4.02)	—	3.88	3.18	0.82	4.97	2.58 (0.31)	3.35
Championship finalists (Tilastopaja)	6 279	7.95	4.53 (3.95–5.11)	—	4.08	6.42	1.57	4.89	2.40 (0.94)	4.13
Long jump, women										
Registry, all ages	20 515	4.17	3.04 (2.70–3.38)	—	2.88	3.40	1.18	2.51	3.58 (0.43)	4.18
Registry, 20–34	1 536	5.43	4.15 (3.03–5.27)	—	3.99	4.88	1.22	4.38	3.42 (1.26)	7.20
US collegiate (TFRRS)	50 068	5.44	2.73 (2.52–2.94)	—	2.86	2.25	0.79	3.81	1.65 (0.23)	2.37
Championship finalists (Tilastopaja)	6 290	6.58	3.71 (3.27–4.16)	—	3.41	4.96	1.46	4.62	1.04 (0.58)	2.84
Triple jump, men										
Registry, all ages	4 237	11.15	4.21 (3.14–5.28)	—	5.18	1.27	0.25	3.39	8.50 (1.60)	7.39
Registry, 20–34	527	13.66	4.89 (2.55–7.23)	—	7.68	−5.28	-0.69	3.94	14.51 (4.48)	10.77
US collegiate (TFRRS)	24 224	14.25	4.57 (4.15–5.00)	—	4.27	5.76	1.35	5.54	2.68 (0.57)	3.56
Championship finalists (Tilastopaja)	4 801	16.85	6.55 (5.45–7.65)	—	5.74	9.50	1.65	5.83	5.54 (1.46)	5.08
Triple jump, women										
Registry, all ages	7 421	9.89	4.15 (3.41–4.90)	—	4.30	3.70	0.86	3.79	5.28 (1.14)	6.02
Registry, 20–34	1 018	11.71	4.43 (3.14–5.72)	—	5.16	0.49	0.10	5.91	3.86 (1.80)	6.30
US collegiate (TFRRS)	32 378	11.68	3.94 (3.64–4.25)	—	3.78	4.60	1.22	4.90	2.34 (0.39)	2.88
Championship finalists (Tilastopaja)	4 467	14.21	5.89 (5.09–6.68)	—	5.86	6.00	1.02	6.88	3.66 (1.22)	5.28

Sprints in seconds, jumps in centimeters, per m/s of wind, signed as gains with tailwind; n is the number of results in athlete-season-event cells with at least two wind-measured marks. Mean mark: all analyzed results (s for sprints, m for jumps), as in Table S4. Per s: the linear gain divided by the mean time. The registry's 20–34 jump rows rest on thin samples. Tailwind and headwind: the separate branch slopes; head/tail: their ratio, with in parentheses the ratio drag-only physics predicts for the same winds and estimator at the athletes' own speeds (100 m and 200 m). 0 to +2 m/s and above +2 m/s: the tailwind slopes of the piecewise model used for H5. Meet FE: the linear slope with month dummies replaced by meet fixed effects. The registry's hurdle events are class-specific (Table S6), so no registry row is given for the hurdles. The external estimates were stable across the registry's robustness variants (cells with at least three results, no exact-zero readings, athlete clustering; for the collegiate sample also without the 2016–2017 seasons, whose short careers the collection under-covers): the 100 and 200 m slopes moved by at most 3 %, and meet fixed effects attenuated the 100 m slopes by 10–20 % in all three samples. External samples: all outdoor results in the six events in the TFRRS profiles of NCAA Division I athletes (shared cleaned build, version 1.2), and the career results of athletes who reached a World Championships, Olympic or other major championship final 1983–2026 (Tilastopaja), after the registry's filters (electronic timing, wind within ±8 m/s, event-specific plausibility bounds; combined-event sub-results excluded). Added after the analysis plan.

Table S4: Effect per second of race time: gain per +1 m/s divided by mean race duration, flat sprints and short hurdles.

Event	Mean time (s)	Gain per +1 m/s (s)	Per second of race time
60 m	9.67	0.059	0.00611
80 m	12.58	0.075	0.00594
100 m	13.09	0.082	0.00629
150 m	20.22	0.081	0.00403
200 m	28.47	0.121	0.00427
60 m hurdles 68 cm	13.16	0.063	0.00478
60 m hurdles 76.2 cm	11.82	0.062	0.00525
80 m hurdles 76.2 cm	13.36	0.072	0.00535
80 m hurdles 84 cm	14.09	0.078	0.00551
100 m hurdles 76.2 cm	15.61	0.108	0.00694
100 m hurdles 84 cm	14.94	0.078	0.00523
100 m hurdles 91.4 cm	15.66	0.084	0.00534
110 m hurdles 91.4 cm	15.90	0.103	0.00646
110 m hurdles 100 cm	15.74	0.096	0.00608
110 m hurdles 106.7 cm	15.53	0.084	0.00540

Straight flat sprints cluster at 0.0059–0.0063 per m/s per second of race time; the bend events (150/200 m) sit about a third lower; hurdle codes bracket the flat rates.

Table S5: Wind effect in the horizontal jumps by age band (gain per +1 m/s; values girls/women / boys/men; 95 % CIs clustered on meet).

Event	Unit	10–11	12–13	14–15	16–17	18–19	20–34
Long jump	cm	—	—	2.89 (2.42–3.35) / 3.36 (2.82–3.89)	3.04 (2.48–3.60) / 3.71 (2.68–4.74)	3.20 (2.40–3.99) / 2.94 (1.58–4.29)	4.15 (3.03–5.27) / 3.11 (1.69–4.52)
Long jump	%	—	—	0.70 (0.58–0.82) / 0.76 (0.62–0.90)	0.62 (0.49–0.76) / 0.66 (0.44–0.89)	0.64 (0.46–0.82) / 0.48 (0.23–0.73)	0.76 (0.53–1.00) / 0.49 (0.23–0.75)
Long jump (zone)	cm	1.83 (1.41–2.24) / 2.39 (1.88–2.90)	1.86 (1.59–2.14) / 2.54 (2.17–2.91)	—	—	—	—
Long jump (zone)	%	0.55 (0.42–0.68) / 0.70 (0.54–0.85)	0.49 (0.41–0.56) / 0.66 (0.56–0.75)	—	—	—	—
Triple jump	cm	—	—	4.18 (3.07–5.29) / 4.20 (2.49–5.91)	4.13 (2.88–5.37) / 4.10 (2.09–6.11)	3.52 (1.68–5.36) / 4.21 (1.67–6.76)	4.43 (3.14–5.72) / 4.89 (2.55–7.23)
Triple jump	%	—	—	0.45 (0.33–0.57) / 0.41 (0.24–0.58)	0.40 (0.28–0.52) / 0.32 (0.15–0.49)	0.31 (0.13–0.49) / 0.32 (0.12–0.52)	0.37 (0.26–0.48) / 0.34 (0.17–0.51)

In each cell the first value is girls/women, the second boys/men. The zone events are contested only in the youngest classes. In centimeters the effect rises with age for girls and women as jump distances and flight times grow; in percent it shows no consistent age direction.

Table S6: Wind effect in short hurdles by class-specific event code (sex-specific rows; flat-sprint reference for the same sex and distance class in that sex's modal age band for the code; 95 % CIs clustered on meet).

Event (hurdle height)	Sex	Modal band	Gain per +1 m/s (95 % CI), s	Flat reference (95 % CI), s	n
60 m hurdles 68 cm	Boys	10–11	0.055 (0.033–0.077)	0.070 (0.065–0.075)	1 214
60 m hurdles 68 cm	Girls	10–11	0.067 (0.048–0.086)	0.064 (0.059–0.068)	2 562
60 m hurdles 76.2 cm	Boys	12–13	0.063 (0.048–0.078)	0.063 (0.059–0.067)	3 435
60 m hurdles 76.2 cm	Girls	12–13	0.062 (0.051–0.073)	0.056 (0.053–0.059)	7 222
80 m hurdles 76.2 cm	Girls	14–15	0.072 (0.051–0.092)	—	2 520
80 m hurdles 84 cm	Boys	14–15	0.078 (0.051–0.104)	—	1 473
100 m hurdles 76.2 cm	Girls	16–17	0.108 (0.089–0.128)	0.082 (0.077–0.087)	1 046
100 m hurdles 84 cm	Boys	14–15	0.133 (0.092–0.174)	0.090 (0.085–0.095)	709
100 m hurdles 84 cm	Women	20–34	0.069 (0.041–0.097)	0.072 (0.067–0.078)	2 842
100 m hurdles 91.4 cm	Boys	16–17	0.084 (0.052–0.115)	0.079 (0.074–0.083)	679
110 m hurdles 91.4 cm	Boys	16–17	0.103 (0.035–0.171)	0.079 (0.074–0.083)	583
110 m hurdles 100 cm	Men	18–19	0.096 (0.036–0.155)	0.078 (0.072–0.083)	656
110 m hurdles 106.7 cm	Men	20–34	0.084 (0.047–0.121)	0.071 (0.066–0.075)	588

The hurdles analysis was added after the main estimations by a post-plan decision and is documented as a post-plan extension. Flat references are the 60 m or 100 m estimate for the same sex in that sex's modal age band; the flat 80 m has too few results at 14–15 for a reference. Codes with fewer than 300 results for one sex have only the other sex's row.

Table S7: Quadratic specification: linear and quadratic wind terms (raw outcome units per m/s: seconds for sprints, meters for jumps, e.g. 0.0328 m = 3.28 cm; SEs clustered on meet).

Event	Linear term (SE)	Quadratic term (SE)	n
60 m	−0.0596 (0.0010)	0.0037 (0.0004)	72 558
80 m	−0.0752 (0.0065)	0.0013 (0.0026)	1 410
100 m	−0.0856 (0.0014)	0.0057 (0.0005)	64 557
150 m	−0.0794 (0.0194)	−0.0052 (0.0109)	328
200 m	−0.1307 (0.0038)	0.0141 (0.0013)	59 246
Long jump	0.0328 (0.0023)	−0.0008 (0.0008)	35 774
Long jump (zone)	0.0194 (0.0012)	0.0015 (0.0004)	44 336
Triple jump	0.0382 (0.0047)	0.0020 (0.0016)	11 658
Triple jump (zone)	0.0342 (0.0060)	0.0030 (0.0018)	5 426

Negative linear terms for sprints mean faster times with tailwind. Positive quadratic terms in the large sprints match the headwind-tailwind asymmetry of Table S2.

Table S8: Headwind/tailwind asymmetry on the 100 m by sex and age band, against drag-only physics at the athletes' own speeds (slopes in seconds per m/s, signed as gains with tailwind; 95 % CIs clustered on meet).

Sex	Age band	n	Mean time (s)	Tailwind slope	Headwind slope	Head/tail ratio (95 % CI)	Drag-only ratio
Girls/women	12–13	2 733	14.80	0.083	0.112	1.35 (0.82–1.88)	1.81
Girls/women	14–15	9 740	13.93	0.071	0.126	1.76 (1.48–2.04)	1.73
Girls/women	16–17	9 062	13.41	0.059	0.112	1.91 (1.59–2.22)	1.68
Girls/women	18–19	4 207	13.04	0.064	0.106	1.66 (1.31–2.01)	1.66
Girls/women	20–34	4 543	12.57	0.057	0.098	1.71 (1.39–2.03)	1.58
Girls/women	All ages	30 392	13.53	0.065	0.114	1.77 (1.58–1.96)	1.73
Girls/women	All ages, meet FE	30 392	13.53	0.058	0.092	1.59 (1.35–1.83)	1.70
Boys/men	12–13	1 755	14.42	0.071	0.141	1.97 (0.87–3.07)	1.92
Boys/men	14–15	7 499	12.79	0.072	0.113	1.56 (1.28–1.85)	1.73
Boys/men	16–17	9 879	12.01	0.062	0.102	1.65 (1.37–1.93)	1.59
Boys/men	18–19	6 030	11.61	0.067	0.093	1.39 (1.03–1.75)	1.55
Boys/men	20–34	8 909	11.36	0.064	0.082	1.27 (1.01–1.52)	1.58
Boys/men	All ages	34 165	12.08	0.066	0.101	1.54 (1.36–1.71)	1.67
Boys/men	All ages, meet FE	34 165	12.08	0.055	0.080	1.46 (1.18–1.74)	1.63

Head/tail ratio: headwind slope over tailwind slope, CI by the delta method. Drag-only ratio: time changes proportional to the change in squared airspeed at each athlete-season's mean speed, imposed on the observed winds and estimated with the same model (with meet fixed effects in the meet-FE rows). Added after the analysis plan.

Table S9: Correction table: tailwind slope over 0 to +2 m/s, and separate tail- and headwind slopes, by event, sex and age band (sprints; jumps event-level, in cm), with 95 % CIs clustered on meet.

Event	Sex	Age band	Tailwind 0 to +2	Tailwind slope per +1 m/s (95 % CI)	Headwind slope per −1 m/s (95 % CI)	n
60 m	Boys/men	10–11	0.057	0.055 (0.046–0.064)	0.088 (0.077–0.098)	7 635
60 m	Boys/men	12–13	0.066	0.056 (0.050–0.062)	0.071 (0.064–0.078)	15 457
60 m	Boys/men	14–15	0.054	0.042 (0.036–0.049)	0.062 (0.051–0.073)	6 955
60 m	Boys/men	16–17	0.046	0.019 (0.004–0.035)	0.064 (0.049–0.080)	570
60 m	Girls/women	10–11	0.055	0.044 (0.036–0.052)	0.085 (0.076–0.094)	9 794
60 m	Girls/women	12–13	0.057	0.047 (0.042–0.051)	0.067 (0.062–0.072)	21 886
60 m	Girls/women	14–15	0.046	0.038 (0.032–0.043)	0.068 (0.062–0.075)	9 320
60 m	Girls/women	16–17	0.030	0.021 (0.007–0.034)	0.091 (0.068–0.113)	389
100 m	Boys/men	12–13	0.082	0.071 (0.041–0.102)	0.141 (0.111–0.170)	1 755
100 m	Boys/men	14–15	0.073	0.072 (0.063–0.081)	0.113 (0.102–0.123)	7 499
100 m	Boys/men	16–17	0.069	0.062 (0.054–0.070)	0.102 (0.094–0.111)	9 879
100 m	Boys/men	18–19	0.079	0.067 (0.058–0.077)	0.093 (0.080–0.107)	6 030
100 m	Boys/men	20–34	0.073	0.064 (0.058–0.071)	0.082 (0.071–0.092)	8 909
100 m	Girls/women	12–13	0.073	0.083 (0.062–0.105)	0.112 (0.091–0.134)	2 733
100 m	Girls/women	14–15	0.073	0.071 (0.063–0.079)	0.126 (0.116–0.135)	9 740
100 m	Girls/women	16–17	0.066	0.059 (0.052–0.066)	0.112 (0.102–0.122)	9 062
100 m	Girls/women	18–19	0.072	0.064 (0.057–0.072)	0.106 (0.092–0.120)	4 207
100 m	Girls/women	20–34	0.055	0.057 (0.050–0.064)	0.098 (0.087–0.109)	4 543
200 m	Boys/men	10–11	0.133	0.107 (0.023–0.191)	0.194 (0.090–0.298)	1 301
200 m	Boys/men	12–13	0.081	0.076 (0.041–0.111)	0.217 (0.174–0.259)	5 443
200 m	Boys/men	14–15	0.079	0.078 (0.059–0.096)	0.188 (0.164–0.212)	7 147
200 m	Boys/men	16–17	0.094	0.074 (0.060–0.088)	0.172 (0.146–0.198)	6 372
200 m	Boys/men	18–19	0.063	0.069 (0.053–0.084)	0.149 (0.124–0.173)	4 158
200 m	Boys/men	20–34	0.070	0.048 (0.026–0.070)	0.151 (0.125–0.178)	4 931
200 m	Girls/women	10–11	0.256	0.099 (0.025–0.173)	0.086 (−0.012–0.185)	1 780
200 m	Girls/women	12–13	0.127	0.105 (0.080–0.130)	0.181 (0.143–0.219)	7 710
200 m	Girls/women	14–15	0.100	0.094 (0.075–0.113)	0.186 (0.162–0.211)	8 640
200 m	Girls/women	16–17	0.091	0.066 (0.049–0.082)	0.208 (0.173–0.243)	5 854
200 m	Girls/women	18–19	0.081	0.065 (0.047–0.082)	0.151 (0.120–0.183)	2 956
200 m	Girls/women	20–34	0.071	0.047 (0.025–0.070)	0.180 (0.143–0.218)	2 954
Long jump	Both	all	—	2.93 (2.58–3.28) cm	3.71 (2.77–4.65) cm	35 774
Long jump (zone)	Both	all	—	2.48 (2.19–2.77) cm	1.41 (0.95–1.87) cm	44 336
Triple jump	Both	all	—	4.63 (3.78–5.48) cm	2.79 (0.78–4.80) cm	11 658
Triple jump (zone)	Both	all	—	4.69 (3.40–5.98) cm	1.90 (−0.25–4.04) cm	5 426

Worked example: a girls' 12–13 100 m run in +1.8 m/s is corrected by $1.8 \times 0.073 = 0.13$ s to still air, using the slope over 0 to +2 m/s. Sprint slopes are in seconds; the four jump rows are in centimeters, pooled across bands. The 60 m rows at 16–17 rest on thin outdoor samples.

Table S10: Meet fixed effects by age band: effect per m/s of wind with month dummies replaced by meet fixed effects (seconds, 95 % CI clustered on meet), sprints.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.055 (0.045–0.066)	0.055 (0.049–0.061)	0.047 (0.040–0.055)	0.030 (0.011–0.048)	—	—
60 m	Girls/women	0.056 (0.049–0.063)	0.049 (0.043–0.054)	0.047 (0.041–0.052)	0.030 (0.012–0.048)	—	—
100 m	Boys/men	—	0.107 (0.072–0.143)	0.071 (0.056–0.085)	0.063 (0.050–0.075)	0.060 (0.046–0.073)	0.060 (0.051–0.069)
100 m	Girls/women	—	0.093 (0.077–0.109)	0.077 (0.066–0.089)	0.065 (0.057–0.072)	0.069 (0.059–0.080)	0.062 (0.054–0.070)
200 m	Boys/men	−0.000 (−0.147–0.146)	0.091 (0.054–0.128)	0.078 (0.054–0.102)	0.063 (0.038–0.088)	0.058 (0.026–0.090)	0.053 (0.033–0.072)
200 m	Girls/women	0.030 (−0.082–0.142)	0.087 (0.060–0.113)	0.080 (0.061–0.100)	0.079 (0.057–0.100)	0.071 (0.047–0.094)	0.059 (0.006–0.111)

Same reporting threshold as Table 3. The 200 m at 10–11 has too few results per meet for the meet-FE estimate to be informative.

Supplementary figures

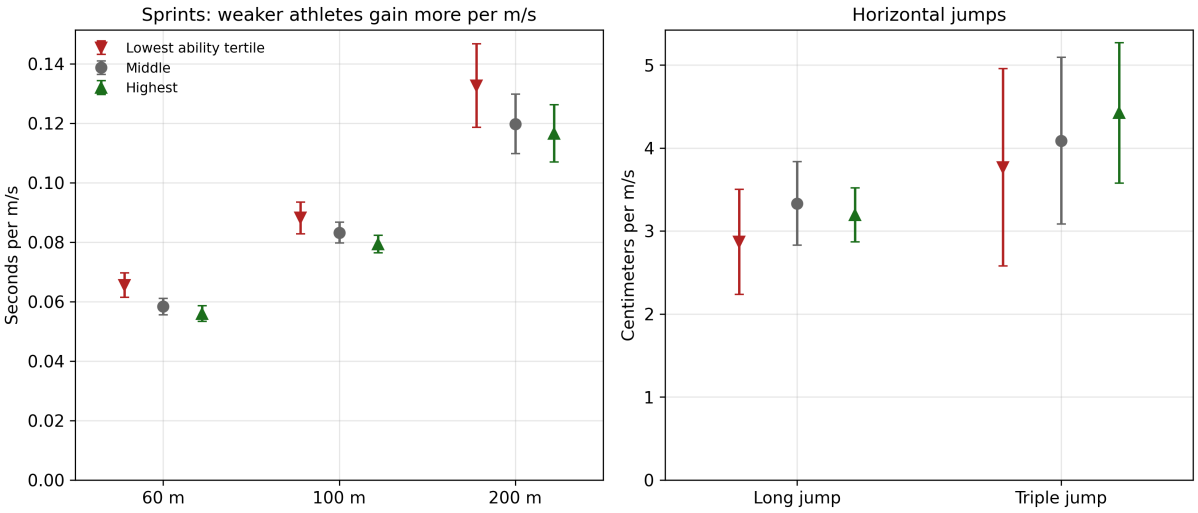


Figure S1: Wind effect by ability tertile: athlete-seasons divided into equal thirds of the wind-adjusted season mean within event × sex × age band (points with 95 % CIs clustered on meet; lowest tertile downward triangles, middle circles, highest upward triangles).

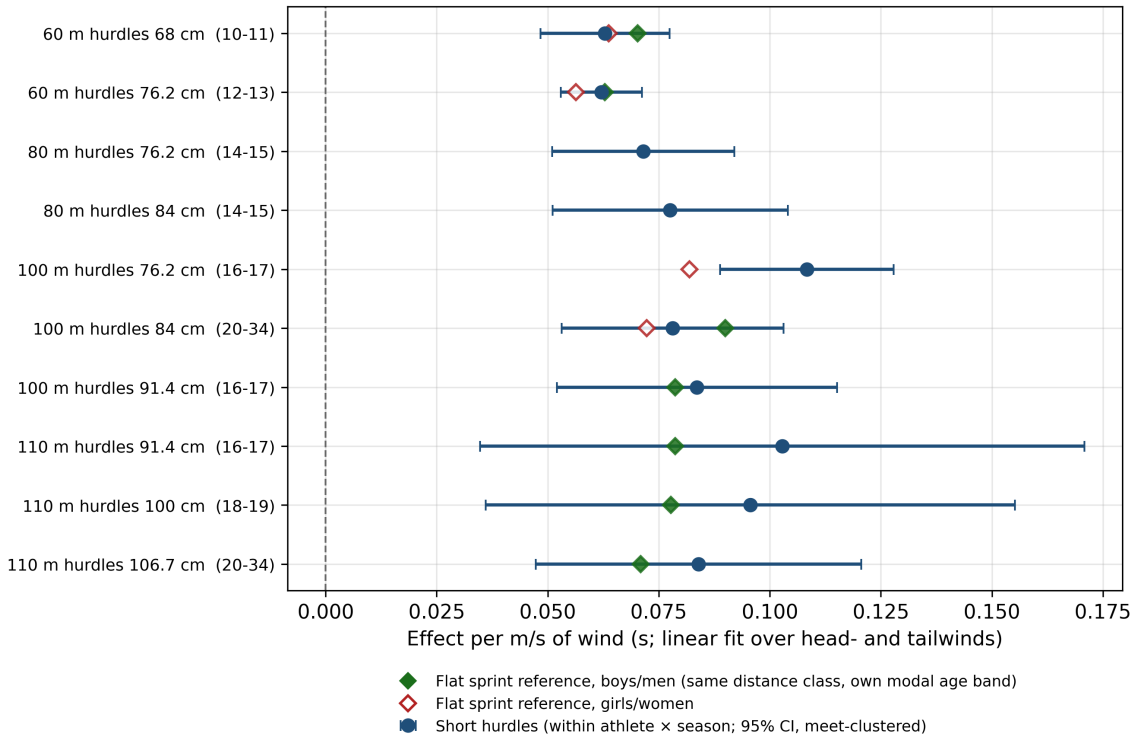


Figure S2: Short hurdles: wind effect per class-specific code (pooled sexes; dots with 95 % CIs clustered on meet) against flat-sprint references for each sex (diamonds; boys/men filled, girls/women open).